

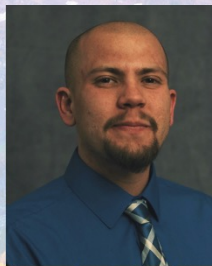


Module 1: Getting Familiar with RC

Meet the User Support Team



Layla
Freeborn



Brandon
Reyes



Andy
Monaghan



Michael
Schneider



John
Reiland



Dylan
Gottlieb



Mohal
Khandelwal

RC Resources

Tech Support

- High Performance Computing
- Data Management
- Cloud Computing
- Secure Research

Human Support

- Training Materials & Workshops
- Consultations & Office Hours
- Help Tickets



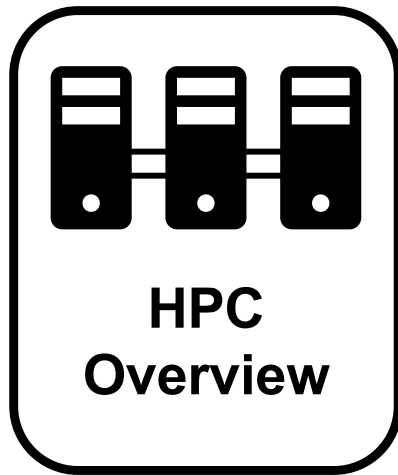
Ask Questions



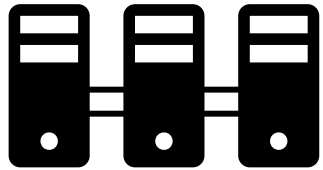
Discuss Ideas



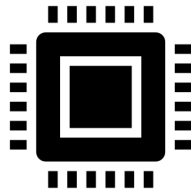
Learning Goals



Learning Goals



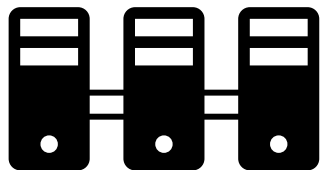
**HPC
Overview**



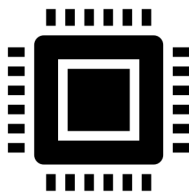
**Nodes &
Partitions**



Learning Goals



**HPC
Overview**



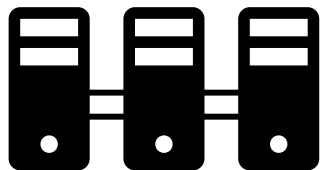
**Nodes &
Partitions**



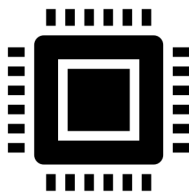
**Data
Storage**



Learning Goals



**HPC
Overview**



**Nodes &
Partitions**



**Data
Storage**



**System
Access**

High Performance Computing



High Performance Computing



High Performance Computing



Scale

vs



Speed



Research Computing
UNIVERSITY OF COLORADO **BOULDER**

08/10/26



Overview



Nodes



Data

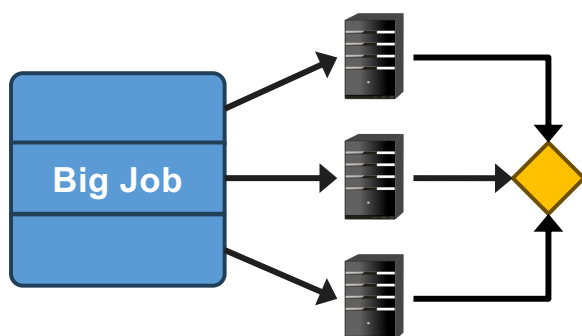


Login

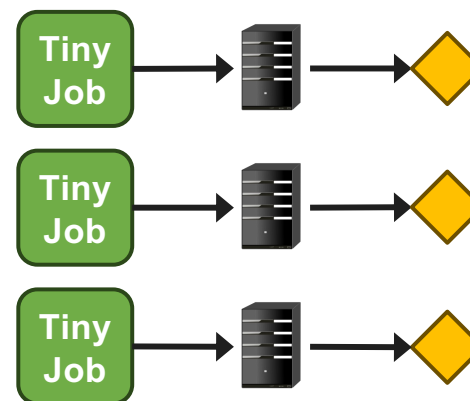
11

Be Boulder.

What can / use HPC for?

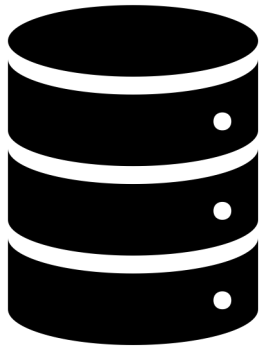


Parallel Jobs

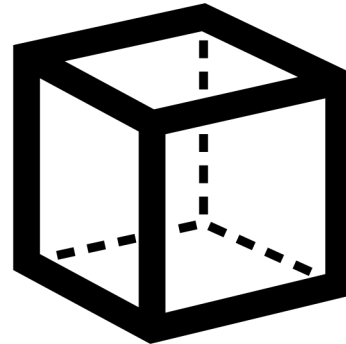


Serial Jobs

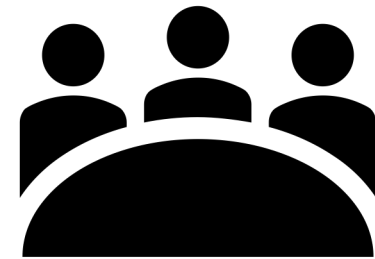
What can / use HPC for?



Big Data



Viz/Rendering



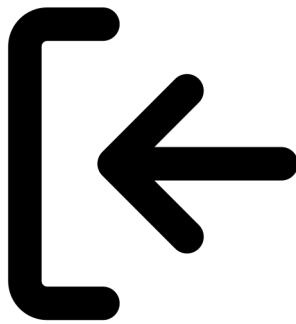
**Shared Work
Environment**

HPC Cluster: Alpine

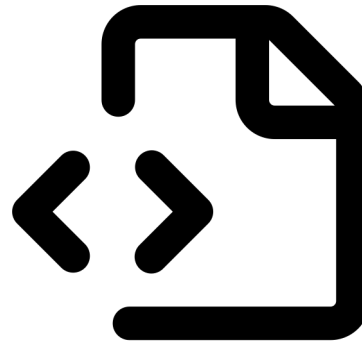


- Heterogeneous cluster
- Hardware provided by CU Boulder, CSU, and AMC
- Access available to CU Boulder, CSU, AMC and RMACC users

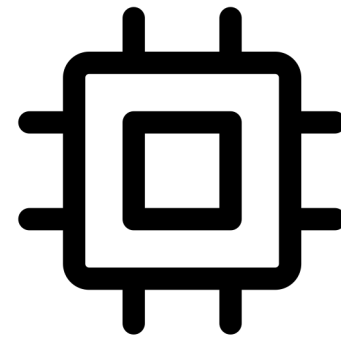
Node Types



LOGIN

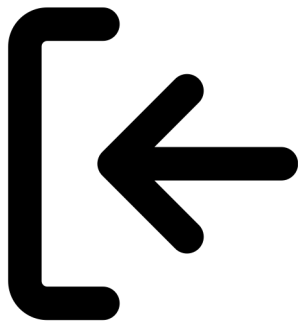


COMPILE



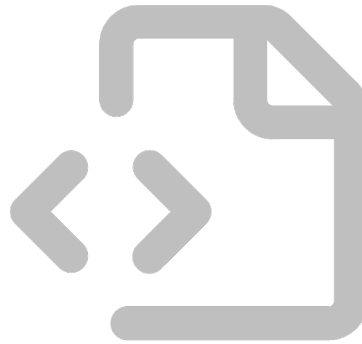
COMPUTE

Node Types

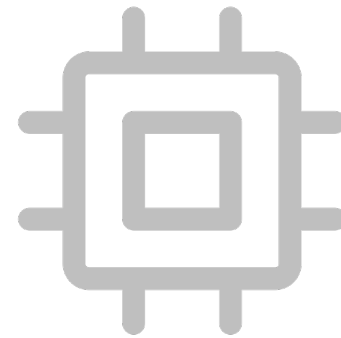


LOGIN

- Entry to system
- View or edit files
- Submit jobs



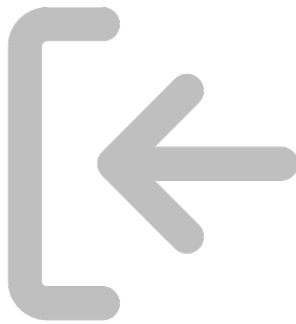
COMPILE



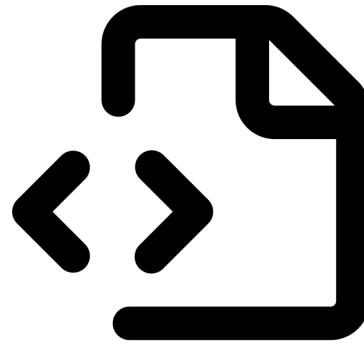
COMPUTE



Node Types

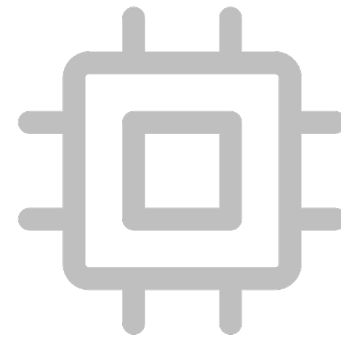


LOGIN



COMPILE

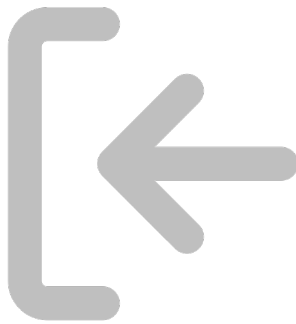
- View or edit files
- Submit jobs
- **Compile code**
- **Install software**



COMPUTE



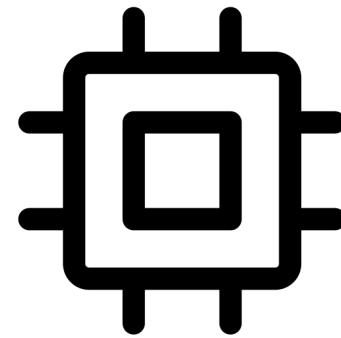
Node Types



LOGIN



COMPILE

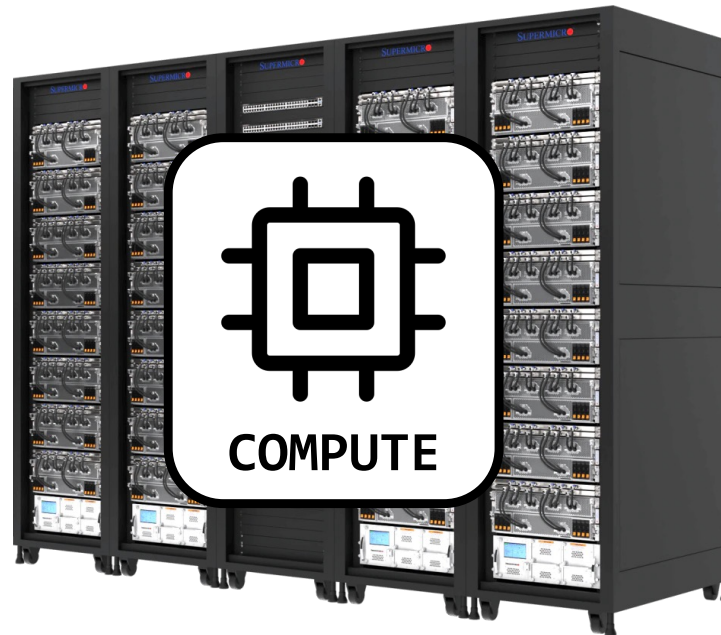


COMPUTE

- Run scheduled jobs
- Handle calculations



Alpine Partitions



Alpine Partitions

acpu

General Usage



Alpine Partitions

acpu

General Usage

amem

High Memory



Alpine Partitions

acpu

General Usage

amem

High Memory



aa100

al40

ah200

artxpro6000

Nvidia GPUs

Alpine Partitions

acpu

General Usage

amem

High Memory



aa100

al40

ah200

artxpro6000

Nvidia GPUs

ami100

AMD GPUs

Data Storage

Core

- Personal Storage
- Includes 3 Directories
 - /home (2 GB)
 - /projects (250 GB)
 - /scratch (10 TB)

Data Storage

Core

- Personal Storage
- Includes 3 Directories
 - /home (2 GB)
 - /projects (250 GB)
 - /scratch (10 TB)

PL

- PetaLibrary
- Tiered Storage
 - Active, Archive
- Requires Funding
- Starts at 1 TB

Core

Home

- Personal data
- Config files
- OnDemand data
- Do not share!

2 GB



Core

Home

- Personal data
- Config files
- OnDemand data
- Do not share!

2 GB

Projects

- Code, Job scripts
- Installed software
- Shared data

250 GB



Core

Home

- Personal data
- Config files
- OnDemand data
- Do not share!

2 GB

Projects

- Code, Job scripts
- Installed software
- Shared data

250 GB

Scratch

- Job data (in/out)
- Shared data
- Deleted after 90 days

10 TB



PetaLibrary

Active

- Performance tier
- Accessible by all nodes
- No file limit



PetaLibrary

Active

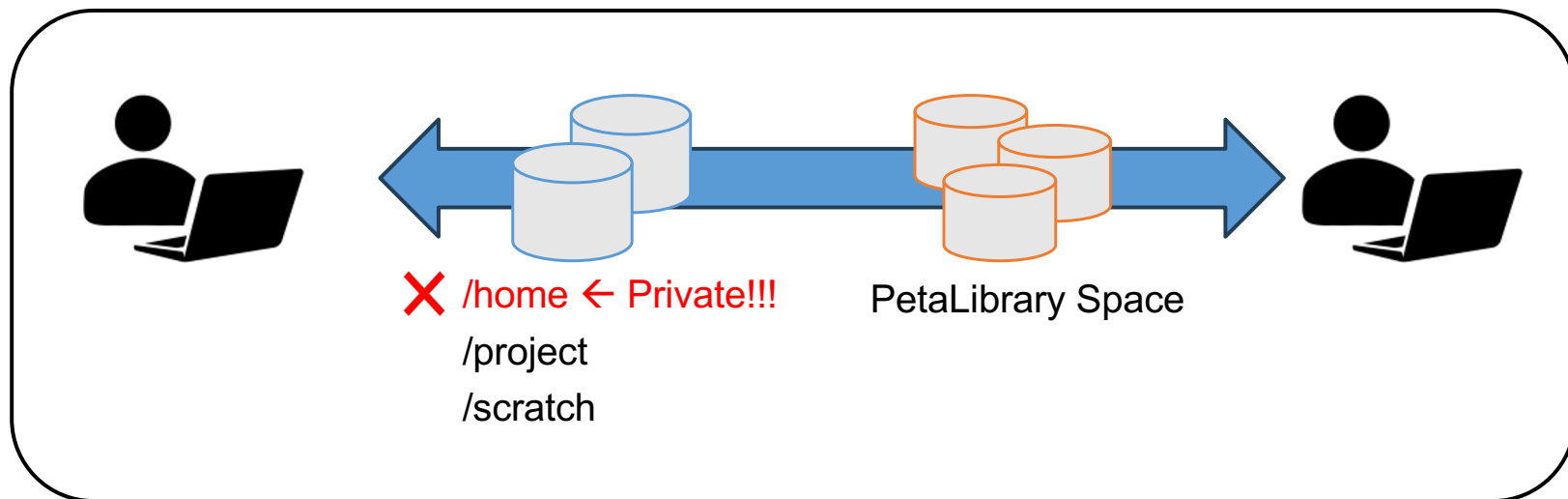
- Performance tier
- Accessible by all nodes
- No file limit

Archive

- Integrity tier
- Accessible by login node
- 10,000 file limit



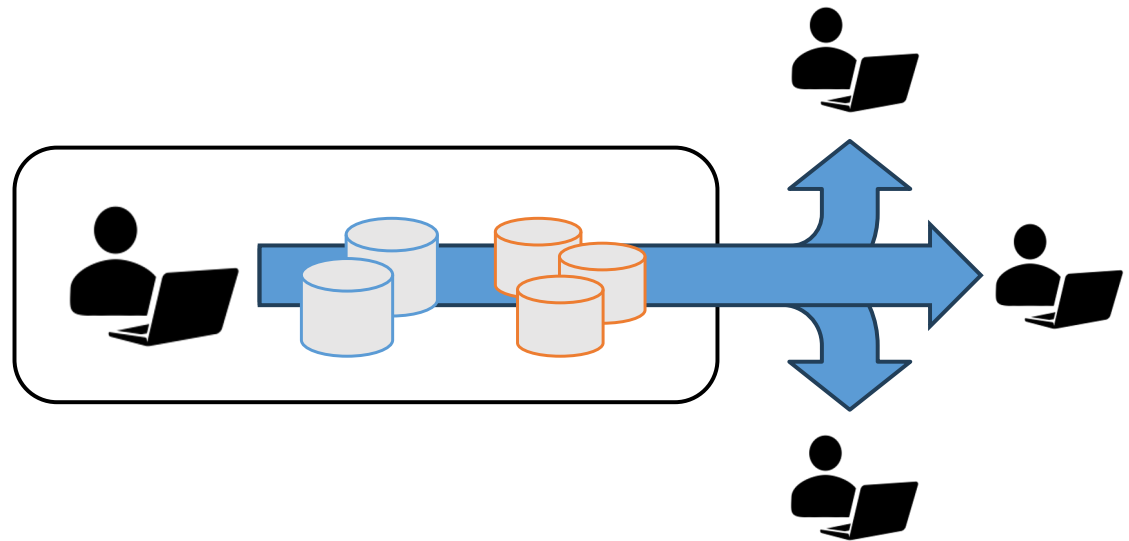
Data Sharing: Within RC



Data Sharing: Outside RC

Tools for data transfers:

- Globus (Recommended)
- Terminal/Command Line:
 - rsync
 - rclone
 - sftp
 - scp



Data Transfer Nodes



DATA (DTN)

- Support data transfers
- Can be selected when using scp, sftp, or ssh transfers
- *dtn.rc.colorado.edu*

Acceptable data storage and use

CURC systems and services **should not be used to store** any data that is US government Classified, nor any Controlled Unclassified Information.

For users requiring storage for sensitive data types, please see the secure research computing resources:

<https://www.colorado.edu/rc/secure-research-computing-resources>

Cloud Computing

- CURC supports AWS, Azure, and GCP
- Alternative or augmentation to HPC
- Interested? Contact rc-help@colorado.edu



How to Access RC Resources?

1. Get an RC account
2. Set up two-factor authentication with Duo
3. Log in via 1) *terminal* or 2) *OnDemand*
4. Create greatness! (responsibly)

Getting an Account

- CU Boulder, CSU users and affiliates:
 - Request an account through the RC Account request portal:
<https://rcamp.rc.colorado.edu/accounts/account-request/create/organization>
- AMC, RMACC users and affiliates:
 - Visit the *AMC* or *RMACC* tab under “Getting a CURC account” in the CURC documentation:
https://curc.readthedocs.io/en/latest/getting_started/logging-in.html

Your RC Account

Access to:

1. Alpine Cluster
2. Core Storage
3. PetaLibrary Storage*
4. OnDemand
5. Approximately 2,000 Service Units (SUs) per month

**If purchased*



Research Computing
UNIVERSITY OF COLORADO BOULDER

08/10/26



Overview



Nodes



Data



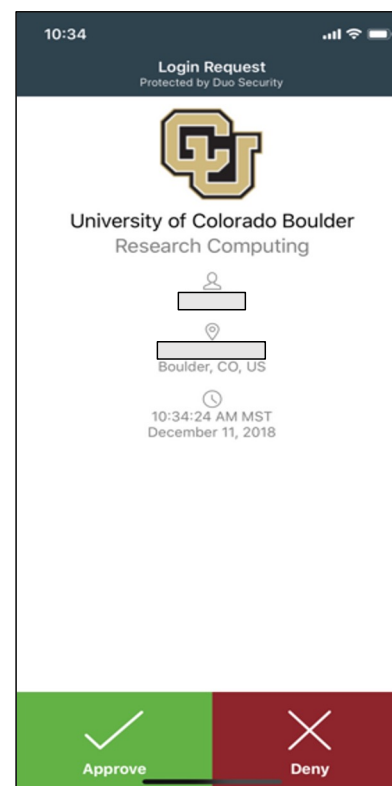
Login

38

Be Boulder.

Duo Authentication

1. Duo smartphone app (recommended)
2. Phone Call/Text is an alternatives



Terminal Access

- Mac or Linux
 - Terminal application
- Windows
 - PuTTY
 - Powershell

```
[user0083@tlogin1 ~]$ pwd  
/home/user0083  
[user0083@tlogin1 ~]$
```

Demo: Logging in via Terminal

To login to an RC login node^{*} :

```
$ ssh <username>@login.rc.colorado.edu
```

Supply your institutional password and your Duo app will alert you to confirm the login

^{*} *Exclusive to CU and CSU accounts. Available on request for AMC. Not available to RMACC.*

Demo: logging in with OnDemand

CURC OnDemand is a browser based, integrated, single access point for all of your HPC resources at CU Research Computing.

- CU Boulder: Visit <https://ondemand.rc.colorado.edu>.
- All other Institutions: Visit <https://ondemand-rmacc.rc.colorado.edu/>

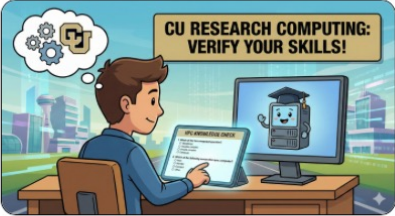
Help! I'm stuck, where do I go?

- **Documentation:** curc.readthedocs.io/
- **Trainings with Center for Research Data and Digital Scholarship (CRDDS):** <https://www.colorado.edu/crdds/>
- **Helpdesk:** rc-help@colorado.edu
- **Consult Hours** (Tuesday 12:00-1:00 in-person or virtually, Thursday 1:00-2:00 virtually)
- Or....

...try CURC's new self-paced training!

Research Computing New User Training

Welcome to the Research Computing New User Training! This course covers the ins and outs of CU Research Computing's (CURC) High Performance Computing (HPC) systems.



Please enter your Research Computing account information:

Do you have an existing RC account?

Institutional email address:

Note: Multiple questions are embedded in this training. It is ok if you don't know the answer to every question! Many of the questions are designed to test for common misconceptions and help you avoid common pitfalls for new users.

https://researchcomputing.github.io/New_User_Async_Training

Helpdesk Tickets

To: rc-help@colorado.edu

Dear Research Computing,

Help! My code won't run!
Help!

Help please,
John

To: rc-help@colorado.edu

Dear Research Computing,

I am running into issues running my Python script. I am using a conda environment called `my_python_env` with the pytorch software, and I am receiving the following error. I am not sure how to troubleshoot. My job ID is 620350. Let me know what I can try!

`sbatch: error: Batch job submission failed: Invalid partition name specified.`

Thanks,
John



Documentation



<https://curc.readthedocs.io/en/latest/>